for $n \neq 1$ $\quad t_{1/2} = \dfrac{2^{n-1}-1}{(n-1)\, k\, [A_0]^{n-1}}$

for $n = 0$ $\quad t_{1/2} = \dfrac{1}{2\, k\, [A_0]^{-1}}$

$t_{1/2} = \dfrac{[A_0]}{2\, k}$

$t_{1/2} \alpha [A_0]$ ———(1)

given

$[A_0] = 0.02M$; $t_{1/2} = 10\,min$

$[A_0] = 0.04M$; $t_{1/2} = ?$

substitute in (1)

$10\ min \alpha\ 0.02M$ ———(2)

$t_{1/2} \alpha\ 0.04M$ ———(3)

$(3) / (2)$

$\Rightarrow \dfrac{t_{1/2}}{10\,min} = \dfrac{0.04\ M}{0.02\ M}$

$t_{1/2} = 2 \times 10\,min = 20\,min$

3. option (b)

$k = A\, e^{-\left(\frac{E_a}{RT}\right)}$

$\ln k = \ln A - \left(\dfrac{E_a}{R}\right)\left(\dfrac{1}{T}\right)$

this equation is in the form of a straight line equatoion

$y = c + m\, x$

a plot of lnk vs $\left(\dfrac{1}{T}\right)$ is a straight line with negative slope

4. option(c)

For a first order reaction

$t_{1/2} = \dfrac{0.693}{k}$

$t_{1/2}$ does not depend on the initial concentration and it remains constant (whatever may be the initial concentration)

$t_{1/2} = 2.5$ hrs

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